Building with AI

ENAS 2170

November 30th, 2026

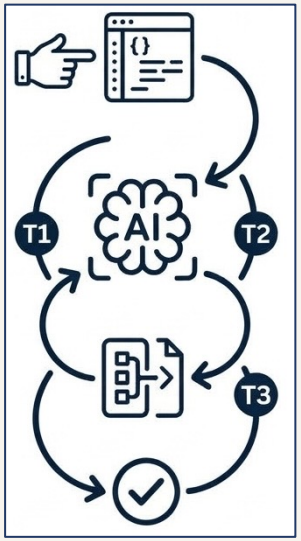
Gaurav Khanna, Ph.D.
AI Executive – Cisco Systems
AI Instructor – Stanford Continuing Studies

Yale College – Class of 1994

AI = Agentic

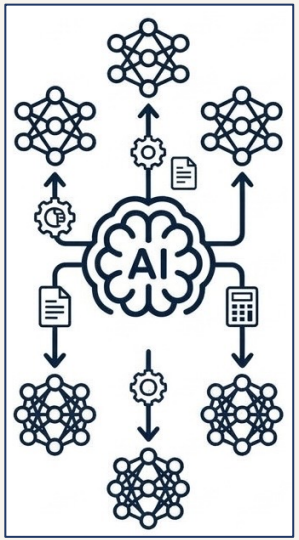
Agentic AI – In a Nutshell

Agentic AI



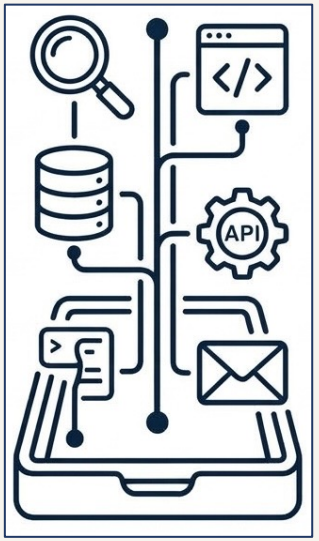
Models

Collection of models selected for the appropriate tasks



Tools

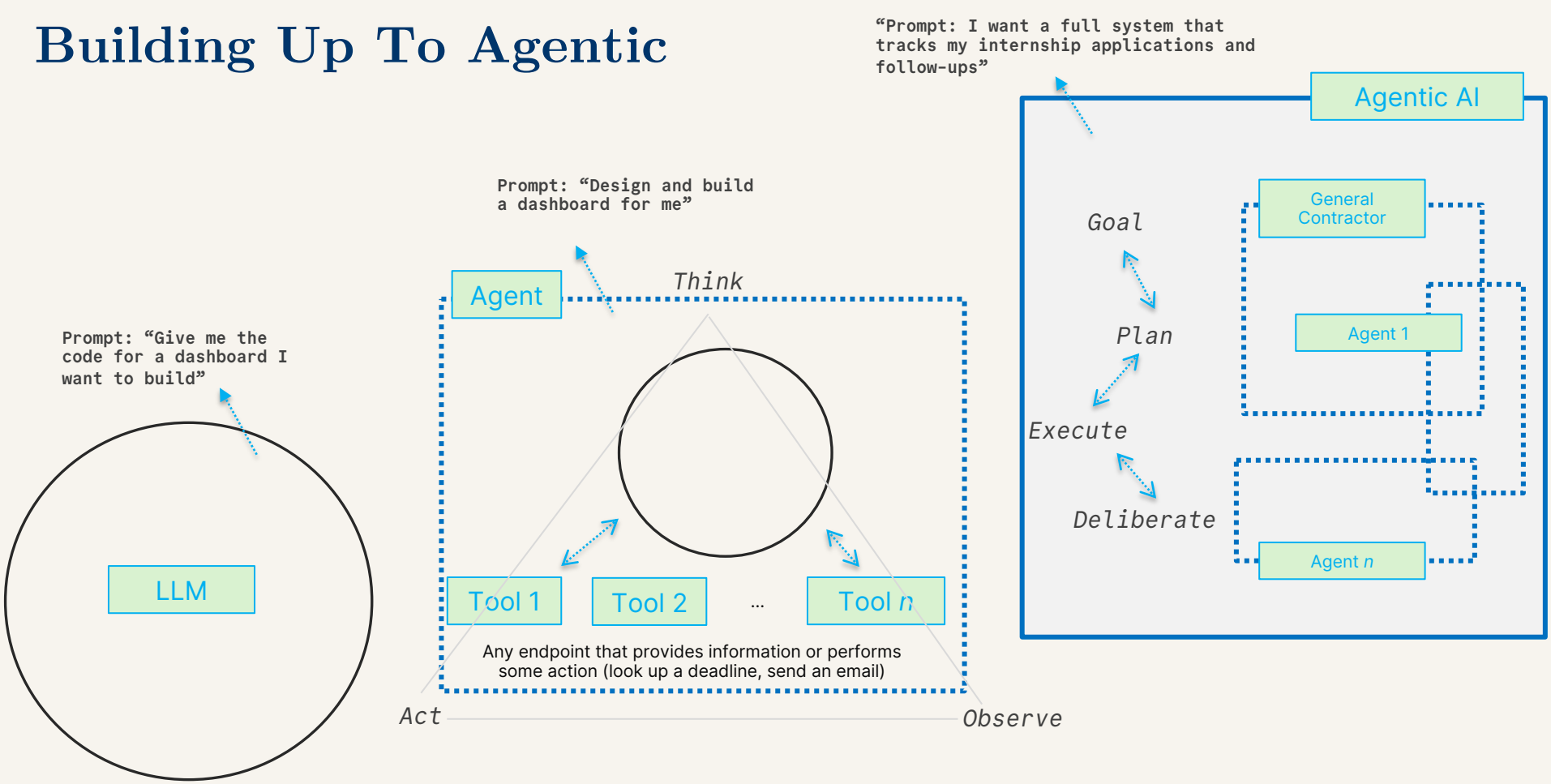
External tools for data and tasks



Harness

Multi-step sequential logic and feedback loops

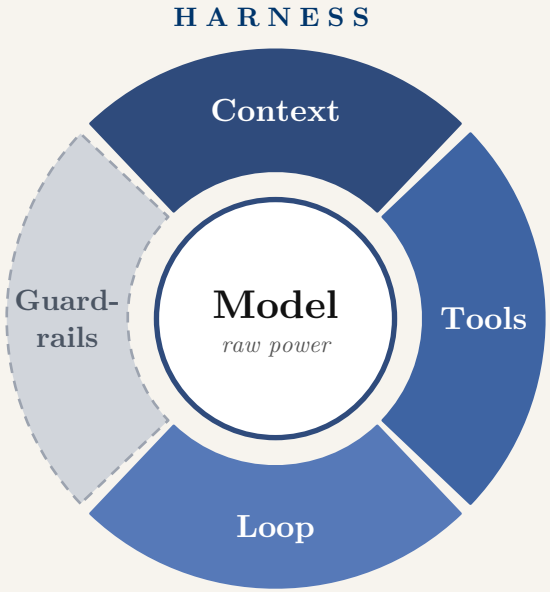
Building Up To Agentic



Agentic systems are built from agentic components
Common elements: LLMs, tools, and iterative processes

The Harness Turns a Model into an Agentic Workhorse

A model is raw power. The harness turns that power into useful, controlled work.



Component	What it Does
Context	What the model sees each turn: instructions, memory, data
Tools	What it can access to act on your behalf
Loop	Think → act → observe → repeat, until done
Guardrails	What it's allowed to do

$$\text{Agent} = \text{Harness}(\text{Model})$$

Like f(x): *the harness is the function; the model is the variable. Swap x, keep f.*

Where the engineering effort went: Prompt engineering → Context engineering → **Harness engineering**

What Are “Skills”?

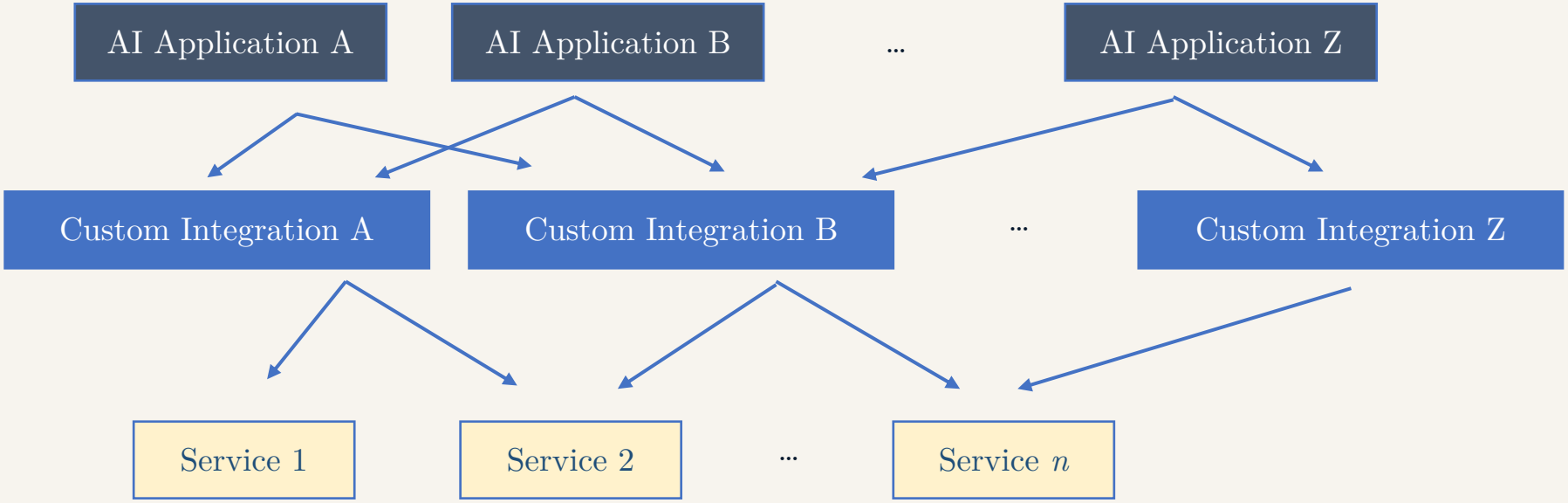
- **Increasingly used as the term for collection of documents to guide LLMs for customization**
- **Important distinction: they persist across multiple sessions and building projects.**

As you engage in a conversation with a coding harness, supporting files and documents get updated (usually by the coding agent itself)



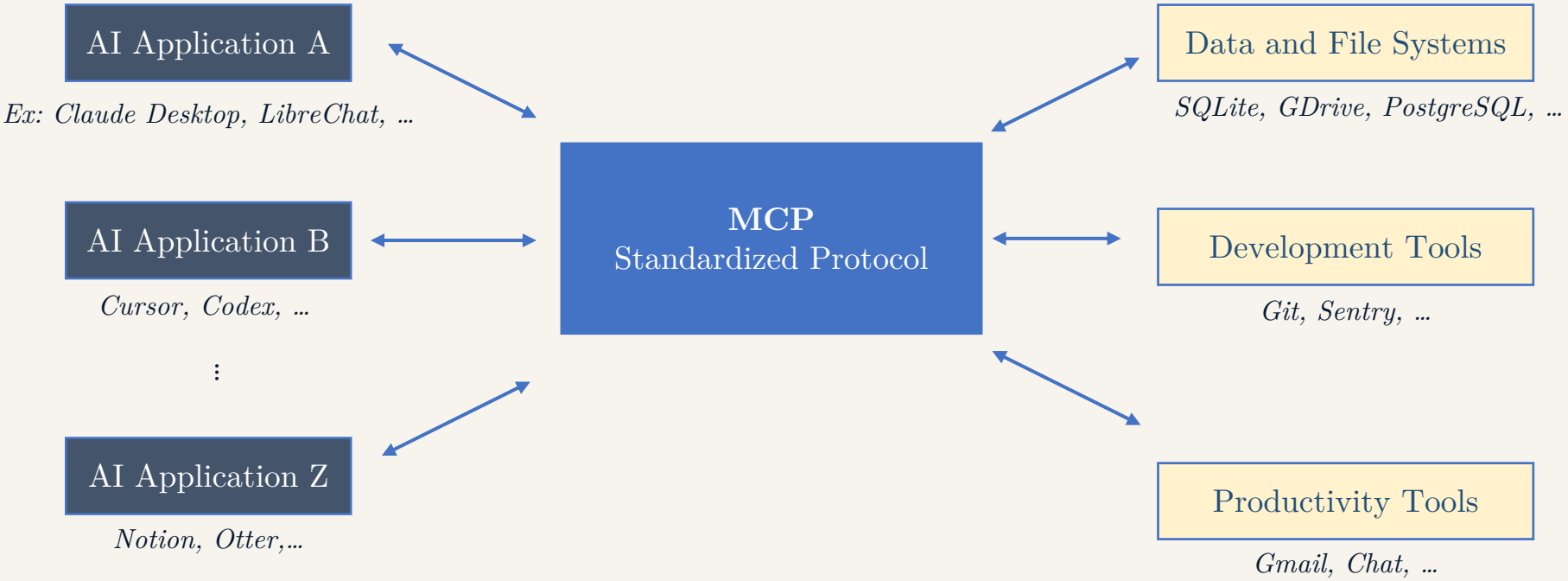
You’ll have an opportunity to build these files during the lab

Prior to MCP: Custom Connections and Integrations



“Full mesh” doesn’t scale; APIs change

MCP: AI's "USB-C Port"

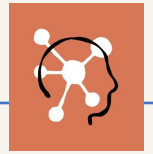


Build one connection, reach any tool.

Only connect MCP servers you trust: each one can see your data and act on your behalf.
Today's server was built for this class [TBD].

Workshop

Lab Overview



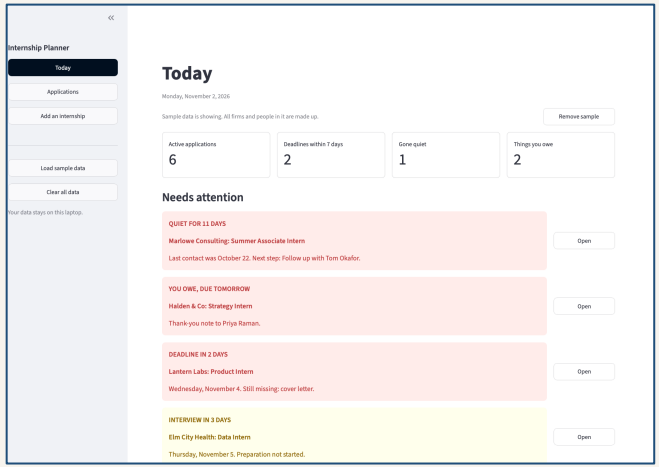






AI Coding Harness
(Claude)

Lab 01:
Internship Discovery and Planner



Lab 02:
Portfolio Tracker

Building With AI Can Be … Counterintuitive!

Building With AI Can Be ... Counterintuitive!



Your education up till now:

learn → practice and get good

→ then do the complex thing



Building with AI:

learn a little → start doing the complex thing

→ learn / ask questions / iterate → keep building

You start building things *before*

you fully understand the underlying tools!

Building With AI Can Be … Counterintuitive

- **There will be elements in the build and tools used which you don't fully understand**
  - Ask questions if you're curious
  - Use the glossary for reference



??!

Starting Prompt

PROMPT

Copy

Build me a simple app called "Knowledge Base" where I can upload PDF documents (for example, product documentation such as Webex guides) and ask questions about them in plain English. Keep all document processing and AI inference running locally on my machine—no cloud AI services or external AI API calls.

Use Python 3.12+, uv for dependencies, and Streamlit for the interface. Give me a PDF upload area, a list of uploaded documents with their processing status, and a question-and-answer area.

Here's how it should work:

- Use PyMuPDF to extract text from each PDF, page by page. Clean up the text and split it into consistent, repeatable chunks while keeping track of filenames and page numbers.
- Use Ollama with nomic-embed-text to create embeddings and ChromaDB to store and search them. Keep document details and chunk mappings in SQLite.
- When I ask something, find the relevant chunks and pass them to Ollama's llama3.2:3b model to generate an answer.
- Instruct the model to answer only from the uploaded documents. Include citations with the filename and page number, and show the retrieved text so I can check the evidence. If the documents don't contain the answer, the model should say so.
- Answer one question at a time; conversation memory isn't needed for the first version.
- Save uploaded PDFs in data/uploads and keep the index between app restarts.
- Create the project in a subfolder named `lab-01-knowledge-base` inside my connected folder. Configure Streamlit to always run on port 8501 (in `.streamlit/config.toml`), so it won't conflict with my other lab apps and they can all run at the same time. Show the address http://localhost:8501 in both guides described below.
- Sample PDFs are in the `Sample PDFs` folder inside my connected folder. Use them to test the app.

Handle repeat uploads sensibly. If the filename and contents are unchanged and the vectors are valid, skip it. If identical contents arrive under a different filename, skip that too. If the filename matches but the contents changed, replace the old version and its indexed content. If processing was incomplete or vectors are missing, repair things when the file is indexed again.

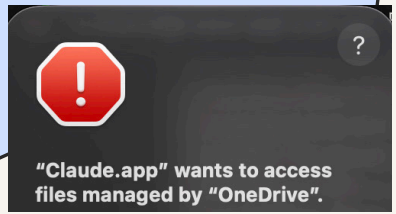
Prompts That Are OK to Ask...

I need help!

Build me a table of the components used in this lab (like Streamlit) and their function

Explain this in plain English...

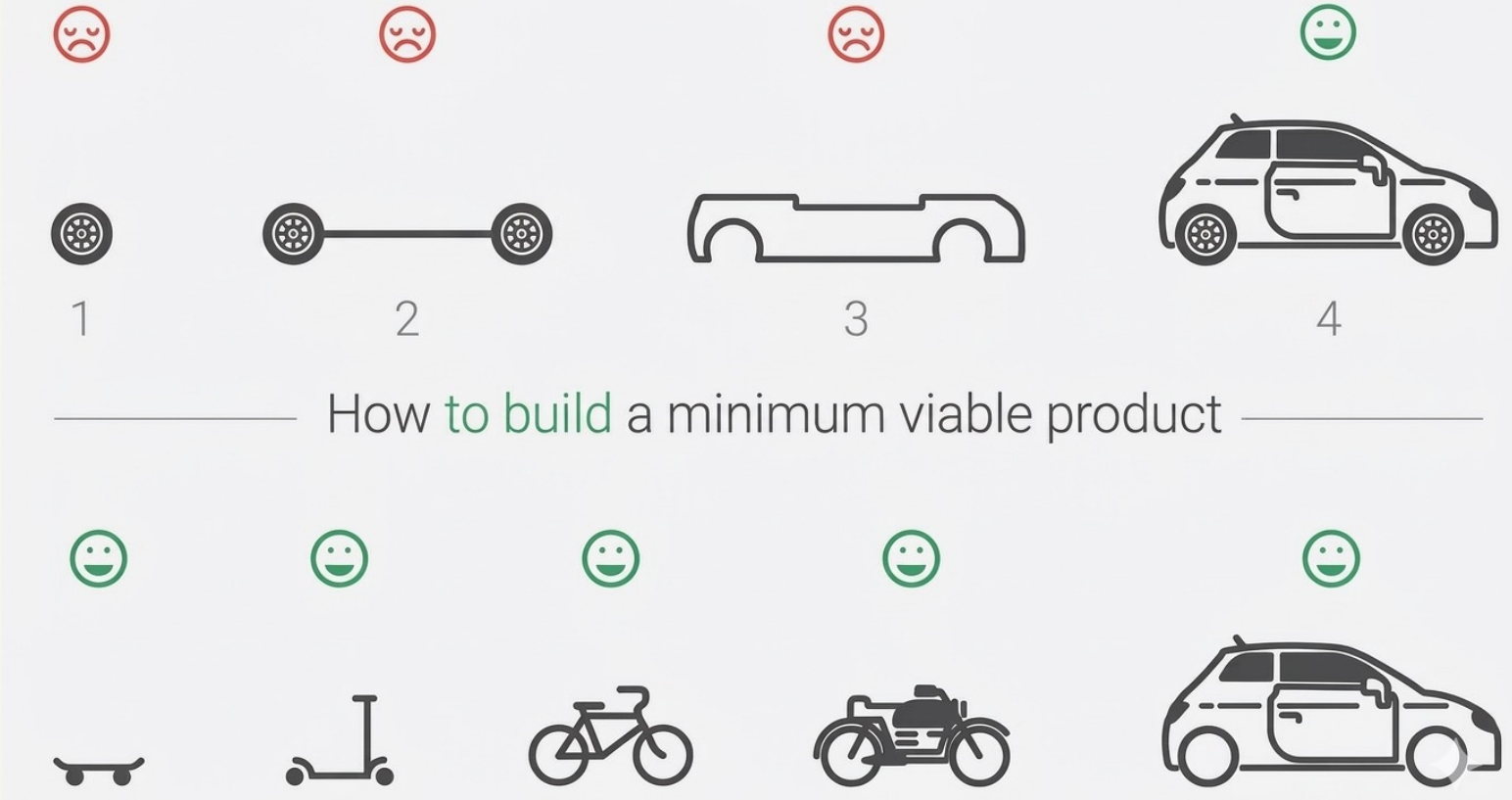
I'm stuck and got this error...



I'm lovin' this!

Give me the same output with the actual terminal commands one-by-one

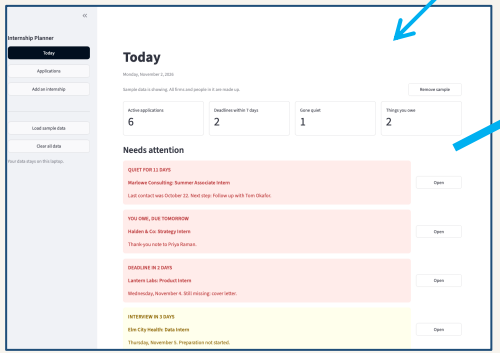
AI is the ultimate "safe space" to ask any question!


1
2
3
4
How to build a minimum viable product

Lab Steps

Group Brainstorm

Build a MVP



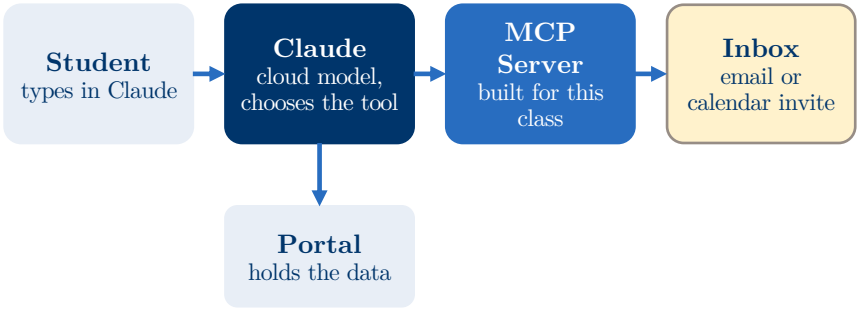
Iterate / Improve / Save a Skills File

Add an agentic component (email reminder or live data)

Making It Do Something: Two Options

Option A: Claude Takes the Action

"Check our planner and email us what's due this week."



Good:

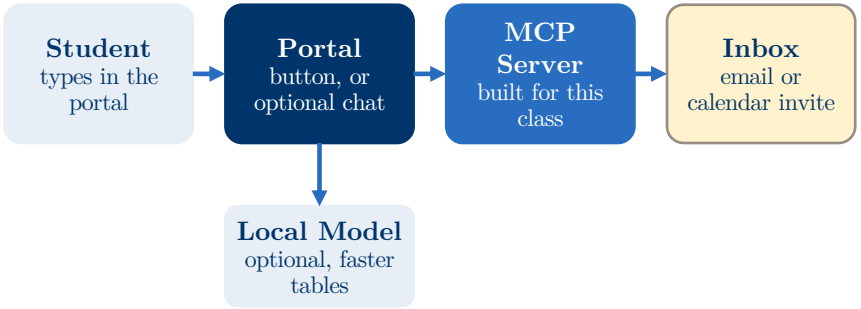
- Nothing extra to install
- Claude is reliable, and students can watch it choose a tool

Consideration:

- Two windows open: the portal and Claude
- Students may ask, "Why build a portal if Claude does the work?"

Option B: The Portal Acts (Leading Choice)

"Remind us about Goldman on Friday," typed into the portal.



Good:

- The portal stands alone; no Claude session needed
- Faster tables build a small agent: Agent = Harness(Model)

Consideration:

- Table leaders install Ollama and a local model ahead of time
- More to build in class. Small models can be less reliable, so they only fill in a fixed format and the portal's code does the sending

Both include an MCP step. **We lean toward B. Does that fit your students?**


“You have taken your first step into a larger world.”